

PROJECT OVERVIEW

1. Project Objective & Workflow

The goal is to build an end-to-end computer vision pipeline that takes an image and outputs a class label (baseball, softball, cricket) alongside a confidence score.

- 1. Dataset Collection & Preprocessing:** Aim for 300-500 images per class.
Collect balanced images for all 3 classes. Resize images to fit model expectations (e.g., 224 $\times$ 224), apply standard normalization, and split data (70% Train, 15% Validation, 15% Test).
- 2. Data Augmentation:** Critical to avoid overfitting on small datasets.
Apply transformations like random horizontal flips, mild rotations, color jittering, and random cropping to help the model learn general visual features rather than background noise.
- 3. Model Selection & Transfer Learning:**
Use pre-trained weights (ImageNet) and replace the final classification head with a dense layer outputting 3 logits.
- 4. Model Training & Evaluation:**
Train using Cross-Entropy Loss and Adam/AdamW optimizer. Measure performance using accuracy, precision, recall, and a confusion matrix.

Models to Use (Choose 1)

Architecture Family	Best Model Candidates	Pros	Cons	Recommendation
Lightweight CNNs	MobileNetV2 / MobileNetV3	Fast training, tiny memory footprint, ideal for mobile or edge deployment.	Lower peak accuracy on fine-grained visual differences.	Best for fast prototyping & edge deployment.
Standard CNNs	ResNet-50 / ResNet-101	Extremely stable training, residual connections prevent gradient issues, strong baseline.	Slightly higher compute requirements.	Solid, dependable baseline.

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Architecture Family	Best Model Candidates	Pros	Cons	Recommendation
Efficient CNNs	EfficientNet-B0 / B2	Optimal balance between accuracy and parameter efficiency.	Sensitive to image resolution changes and hyperparameters.	Best overall accuracy-to-size trade-off.
Vision Transformers	ViT-Base / Swin Transformer	Excellent global context recognition (useful for field layout differences).	High computational cost; overfits easily on small datasets without heavy augmentation.	Use only with large datasets or heavy regularizations.
Zero-Shot Pre-training	CLIP (OpenAI)	Works immediately out-of-the-box without fine-tuning; strong text-image mapping.	Heavy inference footprint; fine-tuning requires specific prompt engineering.	Great for instant baseline comparison.

Important:

Understand what model you chose and why you chose that specific model. What is the advantage does the algorithm that you have chosen have against the other.